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// These are all pin numbers
const int rgbLEDRed = 9;
const int rgbLEDGreen = 10;
const int rgbLEDBlue = 11;

const int buzzer = 3;

const int ledYellowTeam = 4;
const int buttonYellowTeam = 5;

const int ledGreenTeam = 6;
const int buttonGreenTeam = 7;

int yellowTeamButtonPressed = 0;
int yellowTeamScore = 0;

int greenTeamButtonPressed = 0;
int greenTeamScore = 0;

int tick = 0;

// End of pin numbers
String ledColor;
bool buttonPressed = false;

void setup() {
  Serial.begin(9600);
  randomSeed(analogRead(0));

  pinMode(rgbLEDRed, OUTPUT);
  pinMode(rgbLEDGreen, OUTPUT);
  pinMode(rgbLEDBlue, OUTPUT);
  pinMode(ledYellowTeam, OUTPUT);
  pinMode(ledGreenTeam, OUTPUT);

  pinMode(buzzer, OUTPUT);
  noTone(buzzer);
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    pinMode (buttonYellowTeam, INPUT);
    pinMode (buttonGreenTeam, INPUT);

}

void RGB(int Red, int Green, int Blue){
    analogWrite (rgbLEDRed, Red);
    analogWrite (rgbLEDGreen, Green);
    analogWrite (rgbLEDBlue, Blue);
}

void switchLight(int delaySecs){
    int randomNum = random(3);
    if (randomNum == 1 || randomNum == 2) {
        RGB(0, 255, 0);
        tone(buzzer, 2000,1000);
        ledColor = "Green";
    } else if (randomNum == 0) {
        RGB(255, 0, 0);
        tone(buzzer, 1500, 1000);
        ledColor = "Red";
    }
    delay(delaySecs*1000);
}

int buttonDetection (String Color){
    yellowTeamButtonPressed = digitalRead(buttonYellowTeam);
    greenTeamButtonPressed = digitalRead(buttonGreenTeam);
    while (greenTeamButtonPressed == 0 && yellowTeamButtonPressed == 0) {
        tick = tick + 1;

        yellowTeamButtonPressed = digitalRead(buttonYellowTeam);
        greenTeamButtonPressed = digitalRead(buttonGreenTeam);

        if (yellowTeamButtonPressed == 1) {
            if (Color == "Green"){
                yellowTeamScore = yellowTeamScore + 1;
                digitalWrite(ledYellowTeam, HIGH);
                break;
            }
        }
    }
}

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    }else if (Color == "Red"){
        yellowTeamScore = yellowTeamScore - 1;
        tone(buzzer, 1200, 1000);

    }
}

if (greenTeamButtonPressed == 1) {
    if(Color == "Green"){
        greenTeamScore = greenTeamScore + 1;
        digitalWrite(ledGreenTeam, HIGH);
        break;
    }else if (Color == "Red"){
        greenTeamScore = greenTeamScore - 1;
        tone(buzzer, 1200, 1000);
    }
}

if (tick >= 50) {
    Serial.println("Continuing to Next Round...");
    break;
} else if (tick % 49 == 1) {
    Serial.println("Waiting...");
}

delay(100);
}

}

void StartGame(int rounds){
    int currentRound = 1;
    yellowTeamScore = 0;
    greenTeamScore = 0;
    RGB(0, 0, 0);

    while (currentRound <= rounds) {
        ledColor = "";
        digitalWrite(ledYellowTeam, LOW);

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digitalWrite(ledGreenTeam, LOW);

int randomTime = random(3000) + 2001;
tick = 0;

Serial.print("--Round "); Serial.print(currentRound); Serial.println("--");
tone(buzzer,1000,1000);
delay(randomTime);

switchLight(0.5);
buttonDetection(ledColor);
delay(100);

yellowTeamButtonPressed = digitalRead(buttonYellowTeam);
greenTeamButtonPressed = digitalRead(buttonGreenTeam);
while (greenTeamButtonPressed == 1 || yellowTeamButtonPressed == 1) {
    yellowTeamButtonPressed = digitalRead(buttonYellowTeam);
    greenTeamButtonPressed = digitalRead(buttonGreenTeam);
    Serial.println("Waiting for players to lift fingers from buttons...");
    delay(100);
}
RGB(0, 0, 0);
delay(3000);
currentRound = currentRound + 1;
}

Serial.println("---SCORES---");
Serial.print("Yellow: "); Serial.println(yellowTeamScore);
Serial.print("Green: "); Serial.println(greenTeamScore);

if (greenTeamScore > yellowTeamScore) {
    for (int i=0; i<3; i++) {
        RGB(0, 255, 0);
        tone(buzzer, 2000,500);
        delay(500);
        RGB(0, 0, 0);
        delay(500);
    }
}

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} else if (yellowTeamScore > greenTeamScore) {
  for (int i=0; i<3; i++) {
    RGB(255, 220, 50);
    tone(buzzer, 2000,500);
    delay(500);
    RGB(0, 0, 0);
    delay(500);

  }
} else {
  for (int i=0; i<3; i++) {
    RGB(255, 0, 0);
    tone(buzzer, 2000,500);
    delay(500);
    RGB(0, 0, 0);
    delay(500);

  }
}

delay(4000);
}
```

```
void loop() {
  // put your main code here, to run repeatedly:
  StartGame(5);
  RGB(0, 0, 255);
  tone(buzzer, 1700, 4000);
  delay(5000);
}
```